

→ WTW158 CALCULUS STUDY GUIDE 2017

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A. COURSE INFORMATION

1. TEXTBOOK

J Stewart, Calculus Early Transcendentals, 8th Edition.

2. LECTURERS

Module coordinator: Prof AF Harding.

Course administrator: Ms W Conradie.

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Ms E Dreyer	Maths 1-20	3575	ezanda.dreyer@up.ac.za
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Dr H Ohlhoff	Maths 2-22	2851	hohlhoff@gmail.com
Dr Q van der Hoff	Maths 2-11	2785	quay.vanderhoff@up.ac.za

3. CONSULTATION

The consulting hours of the lecturers are displayed on their office doors and on ClickUP. Students may consult lecturers only during the consulting hours as indicated, or by appointment. **This policy also holds before tests and examinations.** This policy aims at encouraging students to plan their work and to work continuously.

4. CALCULATORS

In this course calculators are not allowed at all.

5. ANNOUNCEMENTS

Important announcements are posted on **ClickUP**. This is the primary mode of communicating module matters to students. The announcements may be repeated in lectures and tutorial classes.

It is the student's responsibility to be aware of these announcements.

6. CONTACT TIME

Number of lectures per week: 4

Number of practical classes per week: 1

7. PRACTICAL CLASSES

Generally, each practical session addresses material covered in the lectures of the previous week. The material for the practical is announced on **ClickUP** and in preparation for each practical, you are expected to:

- prepare the theoretical part thoroughly **before the practical**
- do **all the relevant exercises** for the practical **beforehand** and use the practical to sort out any problems.

- do problems assigned in the practical class as well.

It is compulsory to attend one practical class per week.

8. STUDY TIME

A student must devote about 14 hours study time per week to this module. The scheduled contact time is about 7 hours per week, which means that another 7 hours per week of own study time should be devoted to this module.

B. RULES OF ASSESSMENT

1. INSTRUCTIONS

The examination and test instructions of the University of Pretoria must be followed meticulously. A student must also follow all the instructions given by an invigilator when a test is written.

2. SEMESTER TESTS

Semester tests are written during the engineering test weeks. The semester test dates are posted on the UP Website and are also announced on **ClickUP**. Note that there are no sick tests but only one special test (see below) that must be written if a semester test was missed. Students with **special assessment needs (for example, extra time)** must contact the **UP disability centre** for a letter and hand it in to the course administrator before the first test. The material for the test will be announced on **ClickUP**.

3. SPECIAL TEST

If you miss one or both of the semester tests, you **must** write the special test. The date and time will be determined later. The scope of this special test consists of the combined scopes of the first and second semester tests.

4. ClickUP tests, Webassign tasks and Class test.

- ClickUP tests will be written on a weekly basis.
- You will also be expected to complete a number of Webassign tasks throughout the semester.
- Several classtests will be written during the course of the semester.

The material on which the tests and tasks are based, as well as the period in which each test must be completed, are posted on **ClickUP**.

The dates and material for these tests are published on **ClickUP**.

5. ABSENCE FROM TESTS

Absence must be reported to the course administrator within three days of the date of the test and a valid medical certificate must be handed in to the course administrator.

6. ABSENCE FROM EXAMINATIONS

In the case of absence from an examination, the faculty administration should be informed.

7. ENQUIRIES CONCERNING TESTS

All queries concerning the marking of tests must be done within three days after the graded test has been handed back.

8. CHECK YOUR MARKS

After each semester test a complete list of all marks obtained to date will be published. Mistakes, omissions, etc. must be followed up within three days after publication of the list. **No alterations or additions will be made after three days.** Preserve all class and semester tests. It is your only proof that you did indeed write the test.

9. SEMESTER MARK

The semester mark is calculated as follows:

Semester tests	70%
Class tests, ClickUP tests and Webassign tasks	<u>30%</u>
	100%

All the class tests, ClickUP tests and Webassign tasks are used in calculating the semester mark. You must have a semester mark of at least 30% to write the exam.

10. FINAL MARK

The final mark is calculated as follows:

Semester mark	50%
Examination mark	<u>50%</u>
	100%

To pass this course a minimum of 50% is required for the final mark, as well as a subminimum of 40% in the exam.

11. SUPPLEMENTARY EXAMINATION

A student with a final mark between 40% and 49% may write the re-exam. To pass this course a minimum of 50% is required for the re-exam.

12. DISCIPLINARY CASES

The policy of the Department is to refer *all* incidents in which there is a suspicion of irregularity to the Disciplinary Committee of the University.

C. STUDY PLAN

1. USE OF THE STUDY GUIDE

In the study guide, the module is divided into a number of **THEMES** and each theme is further subdivided into **UNITS**. This should provide you with a good overview of the structure of the module. For each lecture unit, a textbook reference for the lecture unit is given under **Source**. In every lecture unit, there are **Learning objectives**. These spell out exactly what you should be able to do in that unit. In addition, you should be able to combine knowledge from different units to solve particular problems.

2. MODULE OBJECTIVES

To have a thorough knowledge of functions, limits, continuity, derivatives and integration and to be able to apply these concepts in problem solving.

3. MODULE OUTCOMES

After completion of this module a student will

- know the definition of, and the properties of linear, power, trigonometric, exponential and logarithmic functions.
- be able to sketch linear, power, trigonometric, exponential and logarithmic functions (also using shifts and translations).
- be able to find limits of a function of a single variable and know what role limits play in finding derivatives.
- be able to identify continuous functions and differentiable functions.
- be able to find derivatives.
- be able to use derivatives to solve problems and to sketch graphs.
- be able to find anti-derivatives.
- be able to evaluate integrals.

4. MODULE STRUCTURE

The subject matter for the course is divided into six themes:

- | | |
|---------------------------------------|-------------|
| 1. Functions | 14 lectures |
| 2. Limits, continuity and derivatives | 9 lectures |
| 3. Derivatives | 7 lectures |
| 4. Application of differentiation | 9 lectures |
| 5. Integration | 6 lectures |
| 6. Vector algebra | 5 lectures |

THEME 1: FUNCTIONS

Unit 1.1 Numbers, intervals and inequalities

Source: Stewart, Appendix A pp A2 - A6

Learning objectives

On completion of this unit you should

1. know the definitions of the following number systems:
the set $\mathbb{N}$ of all natural numbers, the set $\mathbb{Z}$ of all integers, the set $\mathbb{Q}$ of all rational numbers, the set $\mathbb{I}$ of all irrational numbers and the set $\mathbb{R}$ of all real numbers.
2. be able to use set notation.
3. be able to use the definition of the intersection of two sets.
4. be able to use the definition of the union of two sets.
5. be able to use interval notation.
6. know the rules for inequalities and be able to use it.
7. be able to solve inequalities.

Remarks

1. The set $\mathbb{N}$ of all natural numbers is the set $\mathbb{N} = \{1, 2, 3, \dots\}$.
2. Pay special attention to Table [1] on p A4. You must be able to describe an interval in interval notation and in set notation.
3. **Use a graph to solve inequalities where possible.** It is quicker and easier than the algebraic approach.
4. You must know and be able to use the following formula:

$$ax^2 + bx + c = 0 \implies x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Problems for the practical class

Stewart, Exercises A p A9: Numbers: 17, 19, 25, 27, 29, 35, 39, 42, 58, 59, 60.

Unit 1.2 The absolute value

Source: Stewart, Appendix A pp A6 - A8

Learning objectives

On completion of this unit you should

1. be able to write down and use the definition of the absolute value.
2. be able to write down and use the properties of the absolute value.
3. be able to solve inequalities with absolute values.
4. be able to sketch the graph of $y = |x - a|$.

Remarks

1. **When solving inequalities with absolute values, use the theorems given in [6] on p A7.**
2. Omit the triangle inequality on p A8 and Example 9 on p A9.
3. You can also use the following theorem:
If x and y are real numbers, then $|x| \leq |y|$ if and only if $x^2 \leq y^2$.
4. You have to know the following theorem:
If $x \in \mathbb{R}$ then $\sqrt{x^2} = |x|$.

Problems for the practical class

1. Stewart, Exercises A p A9: Numbers: 1, 3, 7, 9, 12, 45, 47, 49, 51, 53, 55, 56.
2. Sketch the graphs of:
 - a. $y = |x - 3|$
 - b. $y = |x^2 - 16|$
 - c. $y = |3x - 2|$

Unit 1.3 What is a function?

Source: Stewart, Section 1.1 pp 10 - 23

Learning objectives

On completion of this unit you should

1. be able to identify a function.

2. be familiar with the four possible ways of representing a function.
3. be able to find the domain of a function.
4. be able to find the range of some functions by using the graph of the function.
5. know how to test whether a curve in the xy -plane is the graph of a function by using the Vertical Line Test.
6. be familiar with piecewise-defined functions.
7. be able to sketch the function $y = |f(x)|$ if the graph of the function $y = f(x)$ is given.
8. be able to write down and use the definition of an even function.
9. be able to write down and use the definition of an odd function.
10. be able to write down and use the definition of an increasing function.
11. be able to write down and use the definition of a decreasing function.

Problems for the practical class

Stewart, Exercises 1.1 p 19: Numbers: 1, 3, 7, 9, 25, 26, 31, 33, 34, 35, 38, 41, 43, 45, 48, 51, 55, 72.

Unit 1.4 Radian measure and trigonometric functions

Source: Stewart, Appendix D pp A24 - A33

Learning objectives

On completion of this unit you should

1. be able to use radian measure.
2. be able to write down and use the definitions of the six trigonometric functions.
3. be able to sketch the graphs of the functions $y = \sin x$, $y = \cos x$ and $y = \tan x$.
4. be able to write down and use some trigonometric identities.
5. be able to solve trigonometric equalities and inequalities.

Remarks

1. Omit Examples 2 and 5.
2. Memorize identities 6 to 13, 15, 16 and 17 on pp A28 - A29.
3. You must be able to calculate all the function values in the table on p A27.

4. You must be able to find the exact trigonometric function values for the angles π , $\frac{\pi}{2}$, $\frac{\pi}{4}$, $\frac{\pi}{3}$ and $\frac{\pi}{6}$ without using a calculator.
5. **We always use radian measure when we work with trigonometric functions.**

Problems for the practical class

1. Stewart, Exercises D p A32: Numbers: 3, 9, 19, 20, 25, 26, 31, 32, 34, 37 (use a calculator), 45, 47, 59, 69, 71, 72, 73, 74, 76.
 2. Find $\sin(75\pi)$ and $\cos\left(\frac{17\pi}{4}\right)$ without using a calculator.
-

Unit 1.5 A catalogue of essential functions

Source: Stewart, Section 1.2 pp 23 - 35

Learning objectives

On completion of this unit you should

1. be able to identify the following functions:
linear functions, power functions, polynomials, algebraic functions, rational functions, exponential functions and logarithmic functions.
2. be able to sketch the graphs of linear functions, power functions, some polynomials, exponential functions and logarithmic functions.

Remark

1. Omit Examples 2, 3 and 4.

Problems for the practical class

Stewart, Exercises 1.2 p 33: Numbers 1, 2, 3, 4, 6, 8, 11, 12, 13.

Unit 1.6 New functions from old functions

Source: Stewart, Section 1.3 pp 36 - 44

Learning objectives

On completion of this unit you should

1. be able to transform a function through translations: vertical and horizontal shifts, stretching, compressing and reflecting.

2. be able to combine two functions f and g to form new functions $f + g$, $f - g$, fg and $\frac{f}{g}$ and be able to determine the domains of these functions.
3. be able to form the composite functions $f \circ g$ and $g \circ f$ for given functions f and g and be able to determine the domain of the new functions.
4. be able to decide what the functions f and g are if the composite function $f \circ g$ is given.

Remarks

1. You should be able to sketch the new transformed function by using the graph(s) of the old function(s).
2. Omit Example 4 on p 39.
3. The functions $y = \sin(Bx)$ and $y = \cos(Bx)$ have period $\frac{2\pi}{|B|}$.

Problems for the practical class

1. Stewart, Exercises 1.3 p 42: Numbers 1, 3, 6, 7, 13, 17, 20, 23, 29, 31, 32, 35, 37, 39, 41, 43, 45, 46, 47, 51.
2. Stewart, Exercises D p A32: Numbers 77 and 81.
3. Sketch the following functions on the interval $[-2\pi, 2\pi]$:
 - a. $f(x) = 2 \sin x$
 - b. $g(x) = \sin(2x)$
 - c. $h(x) = 2 - \sin x$

Unit 1.7 Exponential functions

Source: Stewart, Section 1.4 pp 45 - 55

Learning objectives

On completion of this unit you should

1. be able to write down and use the equation of an exponential function.
2. know that e is the symbol for a very important irrational number.
3. be able to sketch the graphs of the following exponential functions:
 $y = a^x$, $a > 1$ and $y = a^x$, $0 < a < 1$ and $y = e^x$ and $y = e^{-x}$.
4. be able to sketch the graphs of shifted and translated exponential functions.
5. be able to write down and use the laws of exponents.

Remarks

1. You do not have to know how a^x , with x an irrational number, is defined.
If you are interested you can read the explanation on pp 52 - 53.
2. Omit Example 5 on p 53.

Problems for the practical class

Stewart, Exercises 1.4 p 53: Numbers: 1, 3, 13, 14, 15, 17, 18, 19, 21, 30(a) to (c).

Unit 1.8 Inverse functions

Source: Stewart, Section 1.5 pp 55 - 59

Learning objectives

On completion of this unit you should

1. be able to write down and use the definition of a one-to-one function.
2. know how to test whether a function is one-to-one by using the horizontal line test.
3. be able to write down and use the definition of an inverse function.
4. be able to write down the relationship between the domains and ranges of f and f^{-1} .
5. be able to write down and use the cancellation equations of the composites of f and f^{-1} .
6. be able to find the inverse function of a one-to-one function f .
7. be able to explain the relationship between the graphs of f and f^{-1} .

Remarks

1. The definition of a one-to-one function can be written in two ways:
A function $f : A \rightarrow B$ is one to one if $x_1 \neq x_2 \implies f(x_1) \neq f(x_2)$
OR
A function $f : A \rightarrow B$ is one to one if $f(x_1) = f(x_2) \implies x_1 = x_2$
2. The three steps to find a inverse function is given on p 58.

Problems for the practical class

Stewart, Exercises 1.5 p 66: Numbers: 5, 7, 11, 15, 18, 21, 31.

Unit 1.9 Logarithmic functions

Source: Stewart, Section 1.5 pp 59 - 62

Learning objectives

On completion of this unit you should

1. be able to write down and use the definition of the logarithmic function, $y = \log_a x$.
2. be able to write down and use the laws of logarithms.
3. be able to write down and use the definition of the natural logarithmic function, $y = \ln x$.
4. be able to write down and use the properties of the natural logarithm.
5. be able to use the cancellation equations of the composites of $y = \ln x$ and $y = e^x$ to solve equations and inequalities.
6. be able to sketch the graph of the function $y = \ln x$.

Problems for the practical class

Stewart, Exercises 1.5 p 66: Numbers: 23, 25, 36, 38, 39, 41, 51, 52, 54, 55, 56, 57.

Unit 1.10 Inverse trigonometric functions

Source: Stewart, Section 1.5 pp 63 - 66

Learning objectives

On completion of this unit you should

1. be able to write down and use the definitions of the inverse trigonometric functions $y = \arcsin x$, $y = \arccos x$ and $y = \arctan x$.
2. be able to write down and use the properties of the inverse trigonometric functions $y = \arcsin x$, $y = \arccos x$ and $y = \arctan x$.
3. be able to sketch the graphs of the functions $y = \arcsin x$, $y = \arccos x$, $y = \arctan x$.
4. be able to use the cancellation equations of the composites of f and f^{-1} to solve equations.

Remark

Omit the definitions of the other three inverse trigonometric function ([11] on p 69).

Problems for the practical class

THEME 2: LIMITS, CONTINUITY AND DERIVATIVES

Unit 2.1 The tangent and velocity problems

Source: Stewart, Section 2.1 pp 78 - 82

Learning objectives

On completion of this unit you should

1. be able to estimate the slope of a tangent line to a curve by calculation of the limit of the slope of secant lines.
2. be able to explain the concepts "average velocity over an interval" and "instantaneous velocity at time t ".
3. be able to estimate the instantaneous velocity at time t by finding the average velocity over intervals if a distance function is given.

Remark

At a later stage we will develop easier ways of finding instantaneous velocity and the slopes of tangent lines.

Problems for the practical class

Stewart, Exercises 2.1 p 82: Numbers: 1(a), 1(b), 8.

Unit 2.2 The limit of a function

Source: Stewart, Section 2.2 pp 83 - 94

Learning objectives

On completion of this unit you should

1. be able to find the limit of a function at a point if a graph is given.
2. be able to explain what is meant by one-sided limits and be able to interpret these ideas graphically.
3. be able to find $\lim_{x \rightarrow a} f(x)$ by using the one-sided limits $\lim_{x \rightarrow a^+} f(x)$ and $\lim_{x \rightarrow a^-} f(x)$ (only when necessary).
4. be able to explain the informal definition of an infinite limit ($\lim_{x \rightarrow a} f(x) = \infty$ or $\lim_{x \rightarrow a} f(x) = -\infty$) and be able to interpret this idea graphically.

5. be able to find vertical asymptotes of a function.

Remarks

1. It is important to understand the informal idea of a limit as given on p 83.
The formal precise definition is given on p 106. You do not have to know this definition.
2. Examples 4 and 5 are important, they show why we need better methods to find limits.
3. How to find the vertical asymptotes of a function?
Find the domain of the function. If $x = c$ is in the domain of the function then $x = c$ is NOT a vertical asymptote.
If the domain is (a, ∞) , calculate $\lim_{x \rightarrow a^+} f(x)$. If the domain is $(-\infty, a)$, calculate $\lim_{x \rightarrow a^-} f(x)$. If the domain is (a, b) , calculate $\lim_{x \rightarrow a^+} f(x)$ and $\lim_{x \rightarrow b^-} f(x)$.

Problems for the practical class

Stewart, Exercises 2.2 p 92: Numbers: 1, 2, 5, 7, 9, 11, 15, 17, 31, 33, 34, 37, 38.

Unit 2.3 Limit laws

Source: Stewart, Section 2.3 pp 95 - 104

Learning objectives

On completion of this unit you should

1. be able to use the appropriate limit law to find
the limit of the sum of two functions.
the limit of the difference of two functions.
the limit of a constant multiple of a function.
the limit of the product of two functions.
the limit of the quotient of two functions.
2. be able to use the direct substitution property to find the limit of polynomials and rational functions.
3. be able to determine whether a limit exists by looking at the one-sided limits at the point.
4. be able to write down and use the squeeze theorem.
5. be able to use the limits $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$, $\lim_{x \rightarrow 0} \frac{x}{\sin x} = 1$, $\lim_{x \rightarrow 0} \frac{\tan x}{x} = 1$

and $\lim_{x \rightarrow 0} \frac{x}{\tan x} = 1$ to calculate similar limits.

Remarks

1. These laws can be proved by using the formal definition of a limit.
2. The method used in Example 6 is important, remember it!
3. Omit Example 10 on p 101.
4. The “proof” that $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$ appears on p 191 but you do not have to know it.

Problems for the practical class

1. Stewart, Exercises 2.3 p 102: Numbers: 2, 13, 15, 17, 21, 23, 25, 27, 39, 40, 41, 44, 45, 46, 50, 59, 61, 62.
2. Stewart, Exercises 3.3 p 196: Numbers: 39, 40, 43, 44, 45.

Unit 2.4 Continuity

Source: Stewart, Section 2.5 pp 114 - 126

Learning objectives

On completion of this unit you should

1. be able to write down and use the definition of continuity of a function at a point.
2. be able to explain the definition of continuity at a point graphically.
3. be able to determine if a given function is continuous at a point.
4. be able to distinguish between removable, jump and infinite discontinuities.
5. be able to write down and use the definitions of “continuous from the left” and “continuous from the right”.
6. be able to test whether a function is continuous over an interval.
You have to distinguish between the intervals $[a, b]$, (a, b) , $[a, b)$, $(a, b]$, etc.
7. be able to use the theorems on the continuity of combinations of functions (sum, difference, constant multiple, products, quotient).
8. know that the following functions are continuous in every point of their domains
 - polynomials · rational functions · root functions
 - exponential functions · logarithmic functions · trigonometric functions
 - inverse trigonometric functions
9. know that the composite of continuous functions is continuous and be able to use this fact to calculate limits.

Remarks

1. Omit the proofs of the theorems.

2. Omit Examples 2(d), 3 and 10.
3. Omit the Intermediate Value Theorem.

Problems for the practical class

Stewart, Exercises 2.5 p 124: Numbers: 3, 5, 7, 11, 17, 18, 23, 39, 41, 43, 45, 46, 47, 71.

Unit 2.5 Limits at infinity and horizontal asymptotes

Source: Stewart, Section 2.6 pp 130 - 140

Learning objectives

On completion of this unit you should

1. be able to explain what is meant by the following limits:

$$\lim_{x \rightarrow \infty} f(x) = L, \quad \lim_{x \rightarrow -\infty} f(x) = L, \quad \lim_{x \rightarrow \infty} f(x) = \infty, \quad \lim_{x \rightarrow -\infty} f(x) = \infty,$$

$$\lim_{x \rightarrow \infty} f(x) = -\infty, \quad \lim_{x \rightarrow -\infty} f(x) = -\infty.$$

and to interpret the limits graphically.

2. be able to find horizontal asymptotes of a function.

3. be able to find $\lim_{x \rightarrow \pm\infty} \frac{f(x)}{g(x)}$, with $f(x)$ and $g(x)$ polynomials by using the fact that

$$\lim_{x \rightarrow \pm\infty} \frac{1}{x^r} = 0, \text{ with } r \text{ a positive rational number.}$$

Remarks

1. You have to memorize the two limits given in [4] on p 132.
2. Example 4 shows that there may be more than one horizontal asymptote.
3. Omit Example 12. We will sketch functions later.
4. Omit the precise definitions and examples on pp 137 - 140.

Problems for the practical class

Stewart, Exercises 2.6 p 137: Numbers: 3, 5, 7, 15, 17, 19, 21, 23, 24, 25, 27, 31, 33, 35, 37, 39, 41, 47, 49, 51, 52.

Unit 2.6 Derivatives and rates of change

Source: Stewart, Section 2.7 pp 140 - 151

Learning objectives

On completion of this unit you should

1. be able to write down and use the definition of the tangent line to a curve.
2. be able to find the equation of the tangent line to a graph.
3. be able to interpret average velocity and instantaneous velocity as slopes of graphs.
4. be able to find the (instantaneous) velocity of an object if the distance is given as a function of the time.
5. be able to write down and use the definition of the derivative of a function at a point.
6. know that the derivative is the slope of the tangent line to the curve.
7. be able to find derivatives of functions by using the definition.
8. be able to estimate derivatives from tabled data.
9. be able to distinguish between average rate of change over an interval and (instantaneous) rate of change at a point.

Remark

1. There are two formulas to calculate the derivative at a point, [4] and [5] on p 146. You have to know both.
2. Leave Examples 6 and 7.

Problems for the practical class

Stewart, Exercises 2.7 p 148: Numbers: 3, 7, 12, 13, 17, 18, 19, 20, 22, 23, 35, 37, 39, 49, 59.

Unit 2.7 The derivative as a function

Source: Stewart, Section 2.8 pp 152 - 165

Learning objectives

On completion of this unit you should

1. be able to write down and use the definition of the derivative function.
2. be able to find the formula of the derivative function by using the definition.
3. be able to sketch the graph of the derivative function if the graph of the function is given.
4. know the other notations for the derivative.
5. be able to write down and use the definition of a function that is differentiable at a point.
6. be able to write down and use the definition of a function that is differentiable on an interval.
7. know and be able to prove that a differentiable function is continuous in a point and that the converse of the theorem is not true.
8. know the three ways for a function not to be differentiable at a point.
9. be able to illustrate graphically when a function is differentiable in a point.
10. be able to illustrate graphically when a function is not differentiable in a point.
11. know the notation for higher derivatives.

Remark

1. You have to be able to prove Theorem 4 on p 158.
2. Omit Examples 6 and 7.

Problems for the practical class

Stewart, Exercises 2.8 p 160: Numbers: 3, 4, 9, 23, 26, 27, 41, 42, 47, 49, 63.

THEME 3: DERIVATIVES

Unit 3.1 Derivatives of polynomials and exponential functions

Source: Stewart, Section 3.1 pp 172 - 182

Learning objectives

On completion of this unit you should

1. be able to write down and use the rule to find the derivative of a constant function, power functions, the natural exponential function, constant multiple of a function, sum of functions, difference of functions
2. be able to use the rules and formulas in problems.

Remark

You do not have to prove any of the rules in this unit.

Problems for the practical class

Stewart, Exercises 3.1 p 180: All odd numbers from 3 to 31, 33, 36, 45, 49, 55, 56, 57, 58, 59, 61, 67, 71, 74, 81.

Unit 3.2 The product rule and the quotient rule

Source: Stewart, Section 3.2 pp 183 - 190

Learning objectives

On completion of this unit you should

1. be able to write down and use the rule to find the derivative of the product of functions.
2. be able to write down and use the rule to find the derivative of the quotient of functions.
3. be able to use the rules in problems.

Remark

You should be able to prove the product rule by using the appropriate limit theorems. We will discuss another proof, not the one in the textbook, in class.

Problems for the practical class

Stewart, Exercises 3.2 p 188: All odd numbers from 3 to 25, 27, 29, 31, 33, 35 (a), 41, 43, 45, 51, 53, 61(a), 64(a&b).

Unit 3.3 Derivatives of trigonometric functions

Source: Stewart, Section 3.3 pp 190 - 197

Learning objectives

On completion of this unit you should

1. be able to write down the derivatives of the six trigonometric functions.
2. be able to use these derivatives in problems.

Remarks

1. You should be able to prove that $\frac{d}{dx} \sin x = \cos x$ and that $\frac{d}{dx} \cos x = -\sin x$ by using the limit definition.
You may assume that $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$, and do not have to know the proof on p 191.
You must be able to prove that $\lim_{\theta \rightarrow 0} \frac{\cos \theta - 1}{\theta} = 0$.
2. You should be able to prove the formulas for the derivatives of the other four trigonometric functions by using the derivatives of $\sin$ and $\cos$ and the quotient rule.

Problems for the practical class

Stewart, Exercises 3.3 p 196: All odd numbers from 1 to 15, 17, 19, 21, 24, 26(a), 28(a), 29, 33, 35, 37.

Unit 3.4 The chain rule

Source: Stewart, Section 3.4 pp 197 - 207

Learning objectives

On completion of this unit you should

1. be able to use the chain rule to find the derivative of a composite function.
2. be able to write down the derivative of any exponential function $f(x) = a^x$.
3. be able to use the rule and formula in problems.

Remark

You do not have to prove the chain rule.

Problems for the practical class

Stewart, Exercises 3.4 p 204: All odd numbers from 7 to 45, 47, 52, 53, 59, 61, 62, 63, 79, 84(a & b), 96.

Unit 3.5 Implicit differentiation

Source: Stewart, Section 3.5 pp 208 - 217

Learning objectives

On completion of this unit you should

1. know when a function is defined implicitly.
2. be able to differentiate a function given implicitly by using implicit differentiation.
3. be able to write down the formulas for the derivatives of the two inverse trigonometric functions $y = \arcsin x$ and $y = \arctan x$.
4. be able to use the technique and formulas in problems.

Remarks

1. You must be able to prove the formulas for the derivatives of the functions $y = \arcsin x$ and $y = \arctan x$.
2. You do not have to know the derivatives of the other inverse trigonometric functions (p 214).

Problems for the practical class

Stewart, Exercises 3.5 p 215: All odd numbers from 5 to 19, 25, 27, 29, 39, 43, 46, 49, 50, 51, 54, 57, 73, 76.

Unit 3.6 Derivatives of logarithmic functions

Source: Stewart, Section 3.6 pp 218 - 223

Learning objectives

On completion of this unit you should

1. be able to write down and use the formula for the derivative of the logarithmic function $y = \log_a x$.

2. be able to write down and use the formula for the derivative of the logarithmic functions $y = \ln x$ and $y = \ln |x|$.
3. know when and how to use logarithmic differentiation.
4. be able to use the formula and technique in problems.

Remarks

1. You must be able to prove the formulas for the derivatives of a logarithmic functions (p 218).
2. The four formulas given on p 221 is very important.
3. The following result is worth remembering:

$$e = \lim_{x \rightarrow 0} (1 + x)^{\frac{1}{x}} = \lim_{n \rightarrow \infty} (1 + \frac{1}{n})^n .$$

Problems for the practical class

Stewart, Exercises 3.6 p 223: All odd numbers from 3 to 21, 23, 26, 29, 31, 33, 37, 39, 41, 45, 47, 51.

Unit 3.7 Hyperbolic functions

Source: Stewart, Section 3.11 pp 259 - 265

Learning objectives

On completion of this unit you should

1. be able to write down and use the definitions of the hyperbolic functions $y = \sinh x$, $y = \cosh x$ and $y = \tanh x$.
2. be able to sketch the graphs of the hyperbolic functions $y = \sinh x$, $y = \cosh x$ and $y = \tanh x$.
3. be able to write down and use the derivatives of the functions $y = \sinh x$, $y = \cosh x$ and $y = \tanh x$.
4. be able to use the formulas and graphs in problems.

Remarks

1. You should be able to prove the derivatives of the three hyperbolic functions $y = \sinh x$, $y = \cosh x$ and $y = \tanh x$.
2. You do not have to memorize the identities on p 260.
The identities can be proven by using the definitions of the two hyperbolic functions $\sinh$ and $\cosh$.

3. Omit the section on inverse hyperbolic functions.

Problems for the practical class

Stewart, Exercises 3.11 p 264: Numbers: 1, 2, 3, 23, 30, 32, 51.

THEME 4: APPLICATIONS OF DIFFERENTIATION

Unit 4.1 Maximum and minimum values

Source: Stewart, Section 4.1 pp 276 - p 285

Learning objectives

On completion of this unit you should

1. be able to write down and use the definition of an absolute maximum of a function, an absolute minimum of a function, a local maximum of a function, of a local minimum of a function, a critical number of a function.
2. be able to write down and use the Extreme Value Theorem, Fermat's Theorem.
3. know that the converse of Fermat's Theorem is false.
4. be able to find the critical numbers of a function.
5. be able to use the Closed Interval Method to find the absolute extremes of a continuous function on a closed interval.
6. be able to find the absolute and local extremes of a function if the graph of the function is given.

Remarks

1. You do not have to prove any theorems in this unit.
2. It is important to remember that if c is a critical number it does not mean that $f(c)$ is a local extreme. You have to use the tests given in Unit 4.3 to test whether it is a local extreme or not.
3. Omit Example 9.

Problems for the practical class

Stewart, Exercises 4.1 p 283: Numbers: 3, 5, 9, 19, 20, 21, 27, 31, 33, 39, 41, 49, 55, 57, 59, 60, 61, 76.

Unit 4.2 The Mean Value Theorem

Source: Stewart, Section 4.2 pp 287 - 292

Learning objectives

On completion of this unit you should

1. be able to write down Rolle's Theorem and be able to apply it in problems.
2. be able to write down the Mean Value Theorem and be able to apply it in problems.
3. know that a function with a zero derivative on an open interval is a constant function on the interval.
4. know that functions with equal slopes on an interval are at constant distance from each other on the interval.

Remarks

1. You do not have to prove any of the theorems in this unit.
2. Omit Example 6.

Problems for the practical class

Stewart, Exercises 4.2 p 291: Numbers: 3, 5, 8, 13, 17, 21, 27, 29.

Unit 4.3 How the derivative affects the shape of the graph

Source: Stewart, Section 4.3 pp 293 - 304

Learning objectives

On completion of this unit you should

1. be able to write down and use the definition of a function that is increasing on an interval.
2. be able to write down and use the definition of a function that is decreasing on an interval.
3. be able to use the Increasing/Decreasing Test to find the interval(s) on which a function is increasing and the interval(s) on which a function is decreasing.
4. be able to use the First Derivative Test to find the local extremes of a function.
5. be able to write down and use the definition of a function that is concave upward on an interval.

6. be able to write down and use the definition of a function that is concave downward on an interval.
7. be able to use the Concavity Test to find the interval(s) on which a function is concave downward and the interval(s) on which a function is concave upward.
8. be able to write down and use the definition of an inflection point.
9. be able to use the second derivative to find the inflection point(s) of a function.
10. be able to use the Second Derivative Test to find the local extremes of a function.

Remarks

1. The definition of an increasing / a decreasing function is on p 19.
2. You must be able to prove the Increasing / Decreasing Test.
3. Leave Examples 6, 7 and 8 for now. We will do curve sketching in Unit 4.5.

Problems for the practical class

Stewart, Exercises 4.3 p 300: Numbers: 1, 5, 7, 9, 13, 15, 16, 19, 30, 43, 47, 74.

Unit 4.4 Indeterminate forms and L'Hospital's rule

Source: Stewart, Section 4.4 pp 304 - 314

Learning objectives

On completion of this unit you should

1. be able to explain what indeterminate limit forms are.
2. be able to write down the seven different types of indeterminate forms.
3. be able to identify when a limit is of indeterminate form.
4. be able to use the rule of L'Hospital to find certain limits of indeterminate forms.

Remark

1. You always have to write down when a limit is an indeterminate form.

Write it down as: $\lim_{x \rightarrow \infty} \frac{e^x}{x^2}$ (form $\frac{\infty}{\infty}$) = $\lim_{x \rightarrow \infty} \frac{e^x}{2x}$ (form $\frac{\infty}{\infty}$) =

Problems for the practical class

Stewart, Exercises 4.4 p 311: Numbers: 7, 9, 11, 13, 15, 17, 23, 25, 29, 35, 49, 51, 53, 55, 57, 58, 59, 60, 65, 75.

Unit 4.5 Curve sketching

Source: Stewart, Section 4.5 pp 315 – 330

Learning objectives

On completion of this unit you should be able to sketch the graph of a given function.

Remark

When we ask you to sketch the graph of a function you have to follow the guidelines given in this section, you have to

1. find the domain of the function.
2. if possible, find the intercepts of the function with the x -axis and with the y -axis.
3. determine whether the function is even, odd or periodic, or none of these.
4. find the horizontal asymptotes (if any) of the function.
5. find the vertical asymptotes (if any) of the function.
6. find the interval(s) on which the function is increasing and the interval(s) on which the function is decreasing.
7. find the local extremes of the function. (Use number 6.)
8. find the interval(s) on which the function is concave upward and the interval(s) on which the function is concave downward.
9. find the inflection points of the function, and
10. use the information above to sketch the graph of the function.

Remark

Also read examples 6,7 and 8 on pp 298 -299

Problems for the practical class

1. Stewart, Exercises 4.5 p 321: Numbers: 7, 13, 27, 41, 52, 53.
 2. Stewart. Exercises 4.3 p 300: Numbers: 25, 26, 27, 28.
-

Unit 4.6 Optimization problems

Source: Stewart, Section 4.7 pp 331 - 342

Learning objectives

On completion of this unit you should

1. know what is meant by an optimization problem.
2. be able to convert a word problem into a mathematical optimization problem by setting up the function that has to be minimized or maximized.
3. be able to use the methods in this theme to find absolute extreme values.
4. be able to solve optimization problems such as described in the examples.

Remarks

1. The main task in this section is to convert a word problem into a mathematically formulated problem.
2. Making a sketch is invaluable.

Problems for the practical class

Stewart, Exercises 4.7 p 336: Numbers: 1, 3, 5, 9, 16.

THEME 5: INTEGRATION

Unit 5.1 Antiderivatives

Source: Stewart, Section 4.9 pp 350 - 357

Learning objectives

On completion of this unit you should

1. be able to write down and use the definition of an antiderivative of a function.
2. be able to find the most general antiderivative of a function.
3. be able to find a function if the derivative(s) of the function is(are) given.

Remarks

1. You can omit the section about rectilinear motion (pp 354 – 355).
2. Memorize the table on p 352.

Problems for the practical class

Stewart, Exercises 4.9 p 355: All the odd numbers from 1 to 21, 25, 33, 39, 41, 49, 61, 63.

Unit 5.2 The area problem

Source: Stewart, Section 5.1 pp 366 - 378

Learning objectives

On completion of this unit you should

1. be able to estimate the area of a region that lies between the graph of a continuous function, the x -axis and the lines $x = a$ and $x = b$ by using the sum of the areas of rectangles. The rectangles are constructed by using as sample points either the left endpoints or the right endpoints of each subinterval.
2. be able to find an upper and lower estimate of the area of a region that lies between the graph of a continuous function, the x -axis and the lines $x = a$ and $x = b$. The function must be either increasing or decreasing on the interval.

Remark

Omit Example 2

Problems for the practical class

Stewart, Exercises 5.1 p 375: Numbers: 1, 4, 7, 13.

Unit 5.3 The definite integral

Source: Stewart, Section 5.2 pp 378, 379, 385 and 387

Learning objectives

On completion of this unit you should

1. be able to write down and use the definition of definite integral.
2. know what the integrand and limits of integration of a definite integral is.
3. be able to use a Riemann sum to approximate a definite integral.
4. know and be able to use the properties of a definite integral.
5. be able to evaluate integrals by interpreting it in terms of areas.

Remarks

1. Study only Examples 2(a), 4 and 7.

2. If f is integrable on an interval $[a, b]$ and $f(x) \geq 0$ for every $x \in [a, b]$, then $\int_a^b f(x)dx \geq 0$.
In fact, $\int_a^b f(x)dx$ is equal to the area of the region between the graph of f , the x -axis and the lines $x = a$ and $x = b$.
3. If f is integrable on an interval $[a, b]$ and $f(x) \leq 0$ for every $x \in [a, b]$, then $\int_a^b f(x)dx \leq 0$.
In fact, $\int_a^b f(x)dx$ is equal to the negative of the area of the region between the graph of f , the x -axis and the lines $x = a$ and $x = b$.

Problems for the practical class

Stewart, Exercises 5.2 p 388: Numbers: 5(a&b), 7, 33, 35, 37, 39, 49, 50, 57.

Unit 5.4 The Fundamental theorem of Calculus

Source: Stewart, Section 5.3 pp 396 - 398

Learning objectives

On completion of this unit you should know and be able to use the Fundamental Theorem of Calculus, part 2.

Remarks

1. You must be able to prove the Fundamental Theorem of Calculus, part 2.
Do not use the proof in the textbook, we will discuss another proof in class.
2. Only study Examples 5 to 9.

Problems for the practical class

Stewart, Exercises 5.3 p 399: All the odd numbers between 19 and 43, 45, 53, 54, 55.

Unit 5.5 Indefinite integrals

Source: Stewart, Section 5.4 pp 402 - 405

Learning objectives

On completion of this unit you should

1. be able to write down and use the definition of an indefinite integral.

2. be able to evaluate indefinite integrals.

Problems for the practical class

1. Stewart, Exercises 5.4 p 408: Number 1, All the odd numbers from 5 to 17, 31, 34, 40, 45, 46.
2. Stewart, Exercises 5.5 p 418: All the odd numbers from 1 to 71. Omit 47. (**WITHOUT SUBSTITUTIONS!**).

THEME 6: VECTOR ALGEBRA

Unit 6.1 Three- dimensional coordinate systems

Source: Stewart, Section 12.1 pp 792 - 797

Learning objectives

On completion of this unit you should

1. be able to explain what a three-dimensional rectangular coordinate system is.
2. be able to calculate the distance between two points in the three-dimensional space.
3. be able to find the projections of a given point on coordinate planes or coordinate axes.
4. be able to write down the equation of a sphere with given centre and radius.

Problems for the practical class

Stewart, Exercises 12.1 p 796: Numbers: 3, 4, 9, 10, 13, 15, 17.

Unit 6.2 Vectors

Source: Stewart, Section 12.2 pp 798 - 807

Learning objectives

On completion of this unit you should

1. know what a vector is.
2. be able to represent vectors geometrically.
3. understand that a vector has a unique direction and length but not a unique position.
4. be able to represent vectors algebraically.

5. be able to write down a vector with given initial and end points.
6. be able to find the length of a vector.
7. be able to write down and use the definitions of vector addition and scalar multiplication.
8. be able to write down and use the properties of vectors.
9. be able to write down and use the definition of parallel vectors.
10. be familiar with the standard basis vectors in two- and three-dimensional space and be able to express any vector in terms of these.
11. know what is meant by a unit vector.
12. be able to find unit vectors in the same or opposite direction of a given non-zero vector.

Problems for the practical class

Stewart, Exercises 12.2 p 805: Numbers: 3, 4, 7, 9, 15, 19, 21, 22, 23, 26, 29.

Unit 6.3 The dot product

Source: Stewart, Section 12.3, pp 807 - 814

Learning objectives

On completion of this unit you should

1. be able to write down and use the definition of the dot product.
2. be able to determine the dot product of given vectors.
3. be able to write down and use the properties of the dot product.
4. be able to use the dot product to find the angle between two given vectors.
5. be able to use the dot product to determine whether two vectors are perpendicular (orthogonal).
6. be able to determine directional angles and cosines.
7. be able to find scalar and vector projections.

Remark

1. You do not have to do any proofs.

Problems for the practical class

Stewart, Exercises 12.3 p 812: Numbers: 1, 3, 4, 7, 11, 12, 15, 19, 21, 23, 25, 27, 29, 33, 36, 38, 39, 41, 43, 45, 46.

Unit 6.4 The cross product

Source: Stewart, Section 12.4 pp 814 - 819

Learning objectives

On completion of this unit you should

1. be able to write down and use the definition of the cross product of two vectors.
2. be able to find the cross product of two given vectors.
3. be able to find a vector orthogonal to two vectors
4. be able to determine if two vectors are parallel using the cross product.
5. be able to write down and use the properties of the cross product.
6. know the geometric meaning of the cross product.

Remarks

1. Cross products are only defined for three-dimensional vectors.
2. Note that the order of the vectors when calculating the cross product is important.
3. The introduction of a determinant is just to make it easier to calculate the cross product.
4. You do not have to know any proofs.

Problems for the practical class

Stewart, Exercises 12.4 p 821: Numbers: 1, 3, 7, 13, 19, 20, 27, 43.

APPENDIX 1: FORMULAS FOR THE DERIVATIVES OF FUNCTIONS

1. Derivative of a constant function

$$\frac{d}{dx} c = 0, \quad c \in \mathbb{R}$$

2. Derivative of a power function

$$\frac{d}{dx} x^n = nx^{n-1} \quad \text{and} \quad \frac{d}{dx} [f(x)]^n = n[f(x)]^{n-1} \times f'(x) \quad \text{for every real number } n$$

3. Derivative of the natural exponential function

$$\frac{d}{dx} e^x = e^x \quad \text{and} \quad \frac{d}{dx} e^{f(x)} = e^{f(x)} \times f'(x)$$

4. Derivative of any exponential function

$$\frac{d}{dx} a^x = (\ln a) \times a^x \quad \text{and} \quad \frac{d}{dx} a^{f(x)} = (\ln a) \times a^{f(x)} \times f'(x), \quad a > 0$$

5. The derivatives of the trigonometric functions

$$\begin{aligned} \frac{d}{dx} \sin x &= \cos x & \text{and} & \quad \frac{d}{dx} \sin f(x) = \cos f(x) \times f'(x) \\ \frac{d}{dx} \cos x &= -\sin x & \text{and} & \quad \frac{d}{dx} \cos f(x) = -\sin f(x) \times f'(x) \\ \frac{d}{dx} \tan x &= \sec^2 x & \text{and} & \quad \frac{d}{dx} \tan f(x) = \sec^2 f(x) \times f'(x) \\ \frac{d}{dx} \operatorname{cosec} x &= -\operatorname{cosec} x \times \cot x & \text{and} & \quad \frac{d}{dx} \operatorname{cosec} f(x) = -\operatorname{cosec} f(x) \times \cot f(x) \times f'(x) \\ \frac{d}{dx} \sec x &= \sec x \times \tan x & \text{and} & \quad \frac{d}{dx} \sec f(x) = \sec f(x) \times \tan f(x) \times f'(x) \\ \frac{d}{dx} \cot x &= -\operatorname{cosec}^2 x & \text{and} & \quad \frac{d}{dx} \cot f(x) = -\operatorname{cosec}^2 f(x) \times f'(x) \end{aligned}$$

6. The derivatives of the inverse trigonometric functions

$$\begin{aligned} \frac{d}{dx} \arcsin x &= \frac{d}{dx} \sin^{-1}(x) = \frac{1}{\sqrt{1-x^2}} & \text{and} & \quad \frac{d}{dx} \arcsin f(x) = \frac{f'(x)}{\sqrt{1-[f(x)]^2}} \\ \frac{d}{dx} \arctan x &= \frac{d}{dx} \tan^{-1}(x) = \frac{1}{1+x^2} & \text{and} & \quad \frac{d}{dx} \arctan f(x) = \frac{f'(x)}{1+[f(x)]^2} \end{aligned}$$

7. The derivative of logarithmic functions

$$\begin{aligned} \frac{d}{dx} \ln x &= \frac{1}{x}, \quad x > 0 & \text{and} & \quad \frac{d}{dx} \ln f(x) = \frac{f'(x)}{f(x)} \\ \frac{d}{dx} \log_a x &= \frac{1}{x \ln a}, \quad x > 0 \end{aligned}$$

8. The derivatives of the hyperbolic functions

$$\frac{d}{dx} \sinh x = \cosh x$$

$$\frac{d}{dx} \cosh x = \sinh x$$

$$\frac{d}{dx} \tanh x = \operatorname{sech}^2 x$$

APPENDIX 2: RULES OF DIFFERENTIATION

Assume the functions are differentiable.

Derivative of a constant multiple

$$\frac{d}{dx} [cf(x)] = c\left[\frac{d}{dx} f(x)\right]$$

Sum rule

$$\frac{d}{dx} [f(x) + g(x)] = \left[\frac{d}{dx} f(x)\right] + \left[\frac{d}{dx} g(x)\right]$$

Difference rule

$$\frac{d}{dx} [f(x) - g(x)] = \left[\frac{d}{dx} f(x)\right] - \left[\frac{d}{dx} g(x)\right]$$

Product rule

$$\frac{d}{dx} [f(x) \times g(x)] = \left[\frac{d}{dx} f(x)\right] \times g(x) + f(x) \times \left[\frac{d}{dx} g(x)\right]$$

Quotient rule

$$\frac{d}{dx} \left[\frac{f(x)}{g(x)} \right] = \frac{\left[\frac{d}{dx} f(x)\right] \times g(x) - f(x) \times \left[\frac{d}{dx} g(x)\right]}{[g(x)]^2}$$

The chain rule

$$\frac{d}{dx} (f \circ g)(x) = \frac{d}{dx} f(g(x)) = f'(g(x)) \times g'(x)$$

In Leibniz's notation:

If $y = f(u)$ and $u = g(x)$ then

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

APPENDIX 3: STANDARD INTEGRALS

1. Integral of a constant function

$$\int c \, dx = cx + C, \quad c \in \mathbb{R}$$

2. Integral of a power function

$$\int x^n \, dx = \frac{1}{n+1} x^{n+1} + C \quad \text{and} \quad \int [f(x)]^n f'(x) \, dx = \frac{1}{n+1} [f(x)]^{n+1} + C$$

for each real number $n \neq -1$

3. Integral of the natural exponential function

$$\int e^x \, dx = e^x + C \quad \text{and} \quad \int e^{f(x)} f'(x) \, dx = e^{f(x)} + C$$

4. Integral of any exponential function

$$\int a^x \, dx = \frac{1}{\ln a} a^x + C \quad \text{and} \quad \int a^{f(x)} f'(x) \, dx = \frac{1}{\ln a} a^{f(x)} + C, \quad a > 0$$

5. The integrals of the trigonometric functions

$$\begin{aligned} \int \sin x \, dx &= -\cos x + C & \text{and} & \int \sin(f(x)) f'(x) \, dx = -\cos(f(x)) + C \\ \int \cos x \, dx &= \sin x + C & \text{and} & \int \cos(f(x)) f'(x) \, dx = \sin(f(x)) + C \\ \int \sec^2 x \, dx &= \tan x + C & \text{and} & \int \sec^2(f(x)) f'(x) \, dx = \tan(f(x)) + C \\ \int \operatorname{cosec} x \cot x \, dx &= -\operatorname{cosec} x + C & \text{and} & \int \operatorname{cosec}(f(x)) \cot(f(x)) f'(x) \, dx = -\operatorname{cosec}(f(x)) + C \\ \int \sec x \tan x \, dx &= \sec x + C & \text{and} & \int \sec(f(x)) \tan(f(x)) f'(x) \, dx = \sec(f(x)) + C \\ \int \operatorname{cosec}^2 x \, dx &= -\cot x + C & \text{and} & \int \operatorname{cosec}^2(f(x)) f'(x) \, dx = -\cot(f(x)) + C \end{aligned}$$

6. The integrals of the inverse trigonometric functions

$$\int \frac{1}{\sqrt{1-x^2}} dx = \arcsin(x) + C = \sin^{-1}(x) + C \quad \text{and}$$

$$\int \frac{f'(x)}{\sqrt{1-[f(x)]^2}} dx = \arcsin(f(x)) + C = \sin^{-1}(f(x)) + C$$

$$\int \frac{1}{1+x^2} dx = \arctan(x) + C = \tan^{-1}(x) + C \quad \text{and}$$

$$\int \frac{f'(x)}{1+[f(x)]^2} dx = \arctan(f(x)) + C = \tan^{-1}(f(x)) + C$$

7. The integrals of logarithmic functions

$$\int \frac{1}{x} dx = \ln|x| + C \quad \text{and} \quad \int \frac{f'(x)}{f(x)} dx = \ln|f(x)| + C$$

8. The integrals of hyperbolic functions

$$\int \sinh x dx = \cosh x + C \quad \text{and} \quad \int \sinh(f(x))f'(x) dx = \cosh(f(x)) + C$$

$$\int \cosh x dx = \sinh x + C \quad \text{and} \quad \int \cosh(f(x))f'(x) dx = \sinh(f(x)) + C$$

$$\int \operatorname{sech}^2 x dx = \tanh x + C \quad \text{and} \quad \int \operatorname{sech}^2(f(x))f'(x) dx = \tanh(f(x)) + C$$

APPENDIX 4: TECHNIQUE MASTERING EXERCISES

A - ABSOLUTE VALUE (Unit 1.1)

- Write in the form $\begin{cases} \dots & \text{if } x \geq \dots \\ \dots & \text{if } x < \dots \end{cases}$.
a. $|x - 2|$ **b.** $|3 + 6x|$ **c.** $|x^2 - 1|$ **d.** $|x^4 + 4|$
- Solve, using the relevant theorems, and write your answers in interval notation (where possible).
a. $|x - 2| = 0$ **b.** $|x - 2| > 0$ **c.** $|x - 2| < 0$ **d.** $|x - 2| < 1$
e. $|2x| < 7$ **f.** $|2x - 4| > 8$ **g.** $|x - 3| \leq 6$ **h.** $|2x + 5| \geq 6$
i. $|1 - 2x| = 3$ **j.** $|3 - 7x| \leq 6$ **k.** $|5 - 3x| = -3$ **l.** $|-5x - 8| > 12$
m. $|x| = |-3|$ **n.** $|3x| = |2x + 1|$ **o.** $\frac{x - 1}{|2x|} \leq 0$ **p.** $\frac{|2x|}{x - 1} \leq 0$
- Sketch the graphs of the following pairs of functions.
a. $f(x) = x + 4$ and $g(x) = |x + 4|$ **b.** $f(x) = 1 - x^2$ and $g(x) = |1 - x^2|$
c. $f(x) = -x^3$ and $g(x) = |-x^3|$ **d.** $f(x) = 2^x$ and $g(x) = |2^x|$

B - INEQUALITIES

- Solve and write your answers in interval notation.
a. $\frac{3 - x}{-4} \geq$ **b.** $\frac{-4}{3 - x} < 0$ **c.** $\frac{2x - 6}{3x + 9} \geq 0$
d. $\frac{x - 1}{x} < 4$ **e.** $\frac{|-4x|}{3 - x} < 0$ **f.** $\frac{2x - 6}{|3x + 9|} \geq 0$
- Use graphs to solve the inequalities and write your answers in interval notation.
a. $x^2 + 4 > 5$ **b.** $x^2 - 4 > 2 - x$
♣c. $x^6 \geq x^4$ **♣d.** $x^5 < x^7$
♣ - after you have done Unit 1.5

C - FUNCTIONS

1. Determine the domain of f .

Calculate the function value $f(a)$ of f at the given point a .

a. $f(x) = \frac{1}{\sqrt{x-1}}$, $a = 5$ **b.** $f(x) = \frac{3}{|x-4|}$, $a = 1$ **c.** $f(x) = x^4 + x^3 + 1$, $a = 3$

d. $f(x) = \frac{1}{x^2 + x + 1}$, $a = 0$ **♣e.** $f(x) = \frac{1}{\sin x}$, $a = \frac{5\pi}{2}$ **♦f.** $f(x) = e^{\frac{1}{x}}$, $a = \ln 3$

♣ - after you have done Unit 1.4 ♦ - after you have done Units 1.7 and 1.9

a. $f(x) = \frac{1}{\sqrt{x-1}}$, $a = 5$ **b.** $f(x) = \frac{3}{|x-4|}$, $a = 1$ **c.** $f(x) = x^4 + x^3 + 1$, $a = 3$

d. $f(x) = \frac{1}{x^2 + x + 1}$, $a = 0$ **♣e.** $f(x) = \frac{1}{\sin x}$, $a = \frac{5\pi}{2}$ **♦f.** $f(x) = e^{\frac{1}{x}}$, $a = \ln 3$

g. $f(x) = \frac{\sqrt{x}+1}{\sqrt{x}-1}$, $a = 2$ **♣h.** $f(x) = \tan x$, $a = -\frac{\pi}{4}$ **i.** $f(x) = \frac{|x|}{x}$, $a = -1$

j. $f(x) = \frac{x^2 + 2x + 1}{x^2 - 1}$, $a = 0$ **♦k.** $f(x) = \frac{\ln x}{x-1}$, $a = e$ **♦l.** $f(x) = \ln(1 + e^x)$, $a = 0$

♣ - after you have done Unit 1.4 ♦ - after you have done Units 1.7 and 1.9

2. For the given functions f and g , determine $f + g$, $f - g$, fg , $\frac{f}{g}$, $f \circ g$ and $g \circ f$.

a. $f(x) = 2x + 5$, $g(x) = x^2$ **b.** $f(x) = x^2$, $g(x) = \frac{1}{x-1}$

♦c. $f(x) = e^x$, $g(x) = \ln x$ **♣d.** $f(x) = \cos x$, $g(x) = \frac{1}{x}$

e. $f(x) = \sqrt{x}$, $g(x) = x - 1$ **f.** $f(x) = \sqrt{x}$, $g(x) = 2^x$

♣ - after you have done Unit 1.4 ♦ - after you have done Units 1.7 and 1.9

D - TRIGONOMETRIC FUNCTIONS

1. Find, without using a calculator.

a. $\sin\left(\frac{7\pi}{4}\right)$ **b.** $\operatorname{cosec}\left(-\frac{7\pi}{4}\right)$ **c.** $\sin(35\pi)$

d. $\cos\left(-\frac{5\pi}{3}\right)$ **e.** $\sec\left(\frac{5\pi}{3}\right)$ **f.** $\sec\left(\frac{15\pi}{2}\right)$

g. $\tan\left(\frac{7\pi}{6}\right)$ **h.** $\cot\left(-\frac{7\pi}{6}\right)$ **i.** $\cos\left(-\frac{15\pi}{3}\right)$

2. Find, without using a calculator.

- a. $\cos \theta, \tan \theta, \operatorname{cosec} \theta$ and $\cot \theta$ if $\sin \theta = \frac{7}{25}$, $\theta \in \left(\frac{\pi}{2}, \pi\right)$
- b. $\sin \theta, \tan \theta, \sec \theta$ and $\operatorname{cosec} \theta$ if $\cos \theta = -\frac{4}{5}$, $\theta \in \left(\frac{\pi}{2}, \pi\right)$
- c. $\sin \theta, \cos \theta, \cot \theta$ and $\sec \theta$ if $\tan \theta = \frac{5}{12}$, $\theta \in \left(\pi, \frac{3\pi}{2}\right)$
- d. $\sin \theta, \tan \theta, \cot \theta$ and $\operatorname{cosec} \theta$ if $\cos \theta = -1$, $\theta \in \left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$

E - ALGEBRAIC OPERATIONS WITH EXPONENTIALS, LOGARITHMS, ETC

1. Express the following as a power of 2.

- a. $\frac{2}{\sqrt{2}}$ b. $\sqrt[4]{8}$ c. 2.4^{2x}
- d. $2^{x+y} \div 2^{x-y}$ e. $(2^x)^x$ f. $\frac{2^5 \cdot 2^{1/5} \cdot 2^{3/5}}{2^{-1/5}}$

2. Simplify.

- a. $3a^4b^3 \cdot 2a^3b^{-2}$ b. $\frac{12x^{2n+3}}{4x^{n-1}}$ c. $(3x^2y^{-3}z^0)^{-4}$ d. $(3^{x+1})^{x+1}$
- e. $(x^{1/y})^{1-y}$ f. $\sqrt{9x^2}$ g. $(4a^{-2}b^6)^{-\frac{1}{2}}$ h. $\left(\frac{\sqrt{x}}{\sqrt{x-5}}\right)^{-\frac{1}{2}}$
- i. $\frac{7^{-1}x^{1/4}}{3^{-2}y^{-1}} \div \frac{2(3^{-1/2})^2}{(49y)^0}$ j. $\frac{(2^{n-1})^{n-1}}{(2^{n+1})^{n+1}}$ k. $\left(\frac{16a^3}{81a^{-1}}\right)^{-\frac{3}{4}}$
- l. $\frac{12^n \times 8^{n-1} \times 3^{n+1}}{24 \times 6^{n-2} \times 2^n}$ m. $\sqrt[3]{a} \cdot \sqrt{a}$ n. $\frac{(3^n)^{1-n}}{6^{-1} 3^{-n-1}} \div \frac{(3^{1-n})^{n+1}}{2^{-1} 9^{-n-1}}$
- o. $\frac{x^{8n-3}(xy^{1/2})^{-6n}x^3}{(xy)^{-4n}(x^{-1})^n}$ p. $\frac{9^n - 3^{2n+1}}{(3^n)^2 - 3^{n+2} 3^n}$ q. $\frac{5^x - 3^{25}x^{-2}}{5^{x+1} + 3.5^x}$
- r. $(a^{1/2} - b^{3/2})^2$ s. $\frac{a^{-1}}{x^{-1} - a^{-1}}$ t. $\left(\frac{(3y)^{-2}}{3y^{-2}}\right)^{\frac{2}{3}}$

3. Express each logarithm as the sum or difference of simpler logarithms.

- a. $\log\left(\frac{xy}{z}\right)$ b. $\log\sqrt{\frac{x}{y}}$ c. $\log(x^2y^3)$ d. $\log\left(\frac{1}{x^2y}\right)$
- e. $\ln\left(\frac{xy^2}{x+y}\right)$ f. $\ln\sqrt[3]{x^2+1}$ g. $\ln\sqrt[3]{x^2y}$ h. $\ln\sqrt{\frac{x^3}{y}}$
- i. $\log\left(\frac{x}{yz}\right)$ j. $\ln\left(\frac{(x+y)^3}{xy^2}\right)$

4. Express each statement as a single logarithm with coefficient 1.

- a. $3 \ln x$ b. $3 \ln x - \frac{1}{2} \ln y$ c. $\frac{1}{2}(\ln x - 3 \ln y)$
- d. $2 \ln x + \ln(x+y)$ e. $2 \ln x - 3 \ln y + \ln z$ f. $\ln(y-2) + \ln y - 2 \ln x$
- g. $\frac{1}{2}[\ln x - 5(\ln y - 3 \ln z)]$ h. $\frac{1}{2}[(\log x - 5 \log y) - 3 \log z]$
- i. $\log 5 + 3 \log x - 2 \log 4 - \log y$ j. $3 \log(xy) - 2 \log x - \log y$

5. If $\log_a 2 = x$ and $\log_a 3 = y$ express each of the following in terms of x and y .
- a.** $\log_a 27$ **b.** $\log_a \frac{2}{9}$ **c.** $\log_a \sqrt[3]{6}$ **d.** $\log_a \sqrt{12a}$
6. If $f(x) = 2^x$ determine the equations of g , h and k if
- (a) g is symmetrical to f about the x -axis.
 (b) h is symmetrical to f about the y -axis.
 (c) k is symmetrical to f about the line $y = x$.
7. If $\log_a b = 3$, find $\log_b a$.
8. Show that if $y = \frac{r}{1 + ce^{-at}}$ then $t = \frac{1}{a} \ln \frac{cy}{r - y}$.
9. If $f(x) = \log_b x$, find $f(\frac{1}{b})$ and $f(\sqrt{b})$.
10. If $\log_b(\frac{1}{10}) = -\frac{1}{2}$, find b .
11. Simplify
- a.** $4(e^x)^4 + (x^e)^4$ **b.** $\frac{1}{2}e^{2x} \cdot 4e^{3/x}$ **c.** $\frac{14e^{3x}}{7e^x}$
d. $\sqrt{e^x} \cdot \sqrt[3]{e^{-3x}}$ **e.** $\ln(e^{3x})$ **f.** $e^{\ln 7x}$
g. $\ln(9e^x \cdot 10e^{2x})$ **h.** $e^{\ln(\frac{x^2}{y})}$ **i.** $3 \ln x + \ln \frac{1}{x}$
j. $\ln e^2 + e^{-\ln x}$
12. Solve for x without using a calculator.
- a.** $\ln e^{4x} = 12$ **b.** $e^{4x} = 12$ **c.** $\ln 4x = 12$
d. $e^{2 \ln 3} = x$ **e.** $\ln x = 3 \ln 2 - 2 \ln 3$ **f.** $\frac{1}{2} \ln x = 1 - \ln 2$

F - GRAPHS

1. Sketch the graphs of the following functions.
- a.** $f(x) = 2x + 3$ $x \in [-4, 3]$ **b.** $f(x) = |2x - 1|$
- c.** $f(x) = |x| + 1$ **d.** $f(x) = \begin{cases} x^2, & x \leq 1 \\ -x + 2, & x > 1 \end{cases}$
- e.** $f(x) = x^2 - x - 6$ **f.** $f(x) = \sqrt{9 - x^2}$ $x \in [-3, 3]$
- g.** $f(x) = 2 \sin x + 1$ $x \in [-2\pi, \pi]$ **h.** $f(x) = \cos \frac{x}{2}$ $x \in [0, 2\pi]$
- i.** $f(x) = \tan 2x$ $x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

2. Determine the domain of the function and sketch the function

(After you have done Units 1.6 to 1.9)

a. $f(x) = \ln x$ **b.** $g(x) = \ln(-x)$ **c.** $h(x) = \ln |x|$
d. $l(x) = |\ln x|$ **e.** $m(x) = \ln(1+x)$ **f.** $n(x) = 1 + \ln x$

3. Determine the domain of the function and sketch the function

(After you have done Units 1.6 to 1.9)

a. $f(x) = e^x$ **b.** $g(x) = e^{-x}$ **c.** $h(x) = e^{|x|}$
d. $l(x) = |e^x|$ **e.** $m(x) = e^{1+x}$ **f.** $n(x) = 1 + e^x$

G - GRAPHS (After you have done Units 1.7, 1.9, 1.10 and 3.7)

Sketch the graphs of the following pairs of functions on the same set of axes.

1. $f(x) = \sinh x$ and $g(x) = 2 \sinh x + 1$
2. $f(x) = \cosh x$ and $g(x) = \cosh \frac{x}{2}$
3. $f(x) = \tanh x$ and $g(x) = \tanh 2x$
4. $f(x) = \ln x$ and $g(x) = \ln 2x$
5. $f(x) = \ln x$ and $g(x) = \ln(x+2)$
6. $f(x) = \ln x$ and $g(x) = 2 \ln x$
7. $f(x) = e^x$ and $g(x) = e^{2x}$
8. $f(x) = e^x$ and $g(x) = \ln x$
9. $f(x) = \sin x$ and $g(x) = \arcsin x$
10. $f(x) = \cos x$ and $g(x) = \arccos x$
11. $f(x) = \tan x$ and $g(x) = \arctan x$
12. $f(x) = \arcsin x$ and $g(x) = \arcsin(x+1)$

H - LIMITS

Determine the following limits whenever they exist.

- | | | |
|---|--|--|
| 1. $\lim_{x \rightarrow 0} (x + \frac{1}{x})$ | 2. $\lim_{x \rightarrow 1} \frac{x^2 - 1}{x^2 - 2x + 1}$ | 3. $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$ |
| 4. $\lim_{t \rightarrow 0} \frac{t^2 - t}{t}$ | 5. $\lim_{x \rightarrow 2} \frac{\pi}{2}$ | 6. $\lim_{x \rightarrow 0} x $ |
| 7. $\lim_{x \rightarrow e} \ln x^2$ | 8. $\lim_{x \rightarrow \frac{\pi}{2}} \tan x$ | 9. $\lim_{x \rightarrow \frac{\pi}{2}} (1 - \sin x)$ |
| 10. $\lim_{u \rightarrow 2} (u^3 + 4u + 1)$ | 11. $\lim_{x \rightarrow \infty} (x + \frac{1}{x})$ | 12. $\lim_{x \rightarrow \infty} \frac{x^2 + x + 1}{x^3 + 2x^2 + 1}$ |
| 13. $\lim_{x \rightarrow 0^+} \ln x$ | 14. $\lim_{x \rightarrow 3^-} \frac{-1}{\sqrt{3 - y}}$ | 15. $\lim_{x \rightarrow 1^+} \frac{x^2 - 3x + 2}{x^2 - 2x + 1}$ |
-

I - DIFFERENTIATION

Differentiate the following functions.

- | | |
|--|--|
| 1. $f(x) = 2x^4 - 3x^2 + 5x + 2$ | 2. $f(x) = x^{-1/2} + x^{1/2}$ |
| 3. $f(x) = x^2 - 3x^{7/3}$ | 4. $f(t) = t + 1 + \frac{1}{t^2}$ |
| 5. $f(x) = \frac{12}{x} - \frac{4}{x^3} + \frac{1}{x^4}$ | 6. $f(x) = \sqrt[3]{x} + \frac{1}{x^3}$ |
| 7. $f(x) = 3x^2 + \frac{1}{3}x^{-2} + x$ | 8. $f(x) = \sqrt{x} - \frac{1}{\sqrt{x}}$ |
| 9. $f(x) = x^4 - 8x^3 + 2x^2 - x + 1$ | 10. $f(y) = 7y^3 - 5y^2 + 3y - 17$ |
| 11. $f(x) = 9x^{-3} + 2x^{-2} - 14$ | 12. $f(x) = -2x^4 + x^{-2} - 3x^{3/4}$ |
| 13. $f(x) = 4x^{-2} - 7\sqrt{x} + 8x^3 + 5$ | 14. $f(x) = 3x^3 + 2x^2 - x + 1$ |
| 15. $f(x) = x^3 - 3x - 2x^{-4}$ | 16. $f(x) = x + 3 + \frac{4}{x}$ |
| 17. $f(t) = 12t^4 + 3t^3 + 5t^{-2} - 4$ | 18. $f(x) = 4x^3 - 7x^2 + 8x - 6$ |
| 19. $(u) = -3u^{-3} + 4u^2 + \frac{1}{u^2}$ | 20. $f(x) = x\sqrt{x} + \frac{1}{x^2\sqrt{x}}$ |
| 21. $f(x) = (x^3 + 7x^2 + 5)^5$ | 22. $f(x) = \sqrt{x^4 + x^2 + 2}$ |
| 23. $f(t) = \sqrt{t^2 + 2t + 3}$ | 24. $f(x) = x^2(x^3 - 1)$ |
| 25. $f(x) = x\sqrt{1 - x^2}$ | 26. $f(x) = (2x - 4)(3x^2 + 2)$ |
| 27. $f(x) = (\sqrt{x + 1})(x^2 + 1)$ | 28. $f(x) = (x^3 - 3x)^4$ |
| 29. $f(x) = (2x - 3)^3(4x + 2)^2$ | 30. $f(x) = (7x + 3)^2(3x^2 - 14x + 5)$ |

- 31.** $f(y) = (8y^3 - 2)(3y^2 - 5y + 10)^2$ **32.** $f(x) = (2x^3 - 3)^{2/3}$
33. $f(x) = (2x - 3)(3x + 4)$ **34.** $f(x) = (2x^3 - 1)(x^4 + x)$
35. $f(x) = (2x + 1)^2(x^2 + 2)^3$ **36.** $f(x) = \sqrt{3x^2 + 1}$
37. $f(x) = (3x^2 - 2x + 1)^{1/2}$ **38.** $f(x) = (\frac{1}{x} + 3)(x^2 - 5)$
39. $f(t) = (2t - 5)^3$ **40.** $f(x) = (6x - 5)^{-3}$
- 41.** $f(x) = \frac{x^2 + x + 1}{x^2 + 1}$ **42.** $f(x) = \frac{2x + 3}{3x + 2}$ **43.** $f(u) = \frac{2u + 1}{3u - 5}$
44. $f(x) = \frac{x^2 + 5x - 1}{x^2}$ **45.** $f(x) = \frac{(x + 1)(x + 2)}{(x - 1)(x - 2)}$ **46.** $f(x) = \sqrt{\frac{x^2 + 1}{x^2 + 4}}$
47. $f(t) = \frac{t + t^3}{\sqrt{t}}$ **48.** $f(x) = \left(\frac{x + 1}{x - 1}\right)^2$ **49.** $f(x) = \frac{10}{\sqrt{x} + 4}$
50. $f(x) = \frac{8}{4 + x^2}$
- 51.** $f(y) = \frac{4y}{y^2 + 1}$ **52.** $f(x) = \frac{3}{\sqrt{2x + 1}}$ **53.** $f(x) = \frac{3x^2 - 2x + 3}{4x^2 - 5}$
54. $f(x) = \frac{(2x - 4)(3x + 5)}{2x^2 + 7}$ **55.** $f(x) = \frac{2x - 3}{x^2 + 2x}$ **56.** $f(t) = \frac{4 - 2t + 3t^2}{t^2 + 2}$
57. $f(x) = 2x^3 + \frac{2 - x}{x^3}$ **58.** $f(x) = \frac{x - 1}{x + 1}$ **59.** $f(x) = \frac{x^2}{x^2 + 1}$
60. $f(t) = \frac{1}{t^4 - 2t + 1}$
- 61.** $f(x) = \sin(x^2 + 2)$ **62.** $f(x) = \sin(3x)$ **63.** $f(y) = \cos(3y)$
64. $f(x) = \cos(x^4 + 7)$ **65.** $f(x) = \tan(\sin x)$ **66.** $f(t) = \tan(t^2 + 5)$
67. $f(x) = \sin(3x + 2)$ **68.** $f(x) = 2 \sin x - \tan x$ **69.** $f(x) = 1 + x - \cos x$
70. $f(x) = \frac{\sin x}{x}$
- 71.** $f(x) = \tan(2x - x^3)$ **72.** $f(\theta) = \sin(3\theta^2 - 2\theta + 1)$
73. $f(x) = 3 \sin x - 2 \cos x$ **74.** $f(x) = -\cos(\pi x - 1)$
75. $f(x) = x^2 \sin x$ **76.** $f(x) = \cos(\tan x)$
77. $f(x) = 3 \sin(x^2) + 2 \sin(x) - \sin(3)$ **78.** $f(x) = \cos(3x) - \tan(3x)$
79. $f(x) = \sin(2x + 3)$ **80.** $f(t) = \sin(2\pi t)$

- 81.** $f(x) = \sin\left(\frac{1}{\sqrt{x}}\right)$ **82.** $f(x) = \sin\left(\frac{1}{x}\right)$ **83.** $f(x) = \cos(-x)$
84. $f(x) = \tan\pi\left(\frac{1}{2} - x\right)$ **85.** $f(x) = \tan(x^2 + 1)$ **86.** $f(t) = \tan(5t)$
87. $f(x) = x \cos x$ **88.** $f(x) = \frac{\sin x}{1 - x}$ **89.** $f(x) = 4 \sin^2(3x)$
90. $f(x) = \frac{\cos x}{\sin x}$
91. $f(x) = (x^2 + 3) \sin x$ **92.** $f(x) = \sqrt{\sin x}$ **93.** $f(x) = \cos^2(x^3)$
94. $f(x) = x \cos(5x)$ **95.** $f(y) = \sin(2y) \cos(3y)$ **96.** $f(x) = \sin^2(3x) + \cos^2(5x)$
97. $f(x) = x^2 \tan\left(\frac{1}{x}\right)$ **98.** $f(x) = (\sin x - x \cos x)^{-1}$ **99.** $f(x) = \frac{\sin x + \cos x}{\tan x}$
100. $f(x) = \sin^2\left(x + \frac{1}{x}\right) + \cos^2\left(x + \frac{1}{x}\right)$
101. $f(x) = \frac{x^2}{\tan x}$ **102.** $f(x) = \sqrt{\cos(2x)}$ **103.** $f(x) = \sin^2(x - 3)$
104. $f(t) = \cos^3(t^2 - t)$ **105.** $f(x) = \sin x \cos^2 x$ **106.** $f(x) = \sin^{1/3}(2x)$
107. $f(x) = 4 \sin^7(2 - 4x)$ **108.** $f(x) = 2 \sin x + 3 \sin^2 x$ **109.** $f(u) = 3 \cos^3 u + 4 \cos^2 u - 5$
110. $f(x) = 5 \cos^2(x + 2) + 3 \cos(x + 2) - 5$
111. $f(x) = 5x^2 - 3 \tan^2 x + \sin x$ **112.** $f(x) = \frac{\tan x}{1 - \tan x}$ **113.** $f(x) = (\sin(x + 1))^{3/2}$
114. $f(t) = \sin^2 t + \cos^2 t$ **115.** $f(x) = \sin \sqrt{x} + \sqrt{\sin x}$ **116.** $f(x) = (\sin x)(\sin x + \cos x)$
117. $f(x) = 2 \sin x \cos x$ **118.** $f(u) = \frac{1}{\sin u}$ **119.** $f(x) = \cos^2 x \sin x$
120. $f(x) = \sqrt{1 - \sin^2 x}$
121. $f(x) = x + \tan^2 x$ **122.** $f(x) = \frac{x}{\cos x}$ **123.** $f(x) = \tan x \cos x$
124. $f(t) = \sin t \cos t$ **125.** $f(x) = e^{x^2}$ **126.** $f(x) = e^{5x}$
127. $f(x) = (x^2 + 3x)e^x$ **128.** $f(x) = xe^x - e^{-x}$ **129.** $f(x) = e^{x^2} \cdot e^{x+1}$
130. $f(x) = \frac{e^{x^2}}{e^{x-1}}$
131. $f(u) = e^{-u}$ **132.** $f(x) = e^{\frac{2x}{3}}$ **133.** $f(x) = e^{\sqrt{x}}$
134. $f(x) = e^{3x} + 2e^{2x} - 3e^x + 7$ **135.** $f(x) = e^{x^2-2}$ **136.** $f(x) = \frac{1 + e^{2x}}{2 - e^{2x}}$
137. $f(x) = e^{3x-1} - 4e^{-x}$ **138.** $f(t) = \cos(e^t)$ **139.** $f(x) = 3e^{2x} - 4e^x + 1$
140. $f(x) = e^{3 \cos(2x)}$

- 141.** $f(x) = e^{-2x} + 4e^{-3x} + 7$ **142.** $f(x) = e^{2x+1}$ **143.** $f(x) = \frac{1}{2}e^{2x}$
144. $f(x) = e^{\sin x}$ **145.** $f(x) = e^{2x}$ **146.** $f(t) = 2te^t$
147. $f(x) = \frac{1}{1 - e^{-x}}$ **148.** $f(x) = \frac{e^{-x}}{x}$ **149.** $f(x) = x^2e^{-x}$
150. $f(x) = e^{-\frac{1}{x^2}}$
151. $f(x) = e^{\sqrt{x^2+1}}$ **152.** $f(x) = \frac{e^x - e^{-x}}{2}$ **153.** $f(x) = e^{2x} \cos(3x)$
154. $f(x) = e^{\cos(4x)}$ **155.** $f(x) = x^2 \cdot 2^x$ **156.** $f(x) = 3^{5x}$
157. $f(x) = x^4 + 4^x$ **158.** $f(x) = 9^{-x}$ **159.** $f(x) = \tan(5^x)$
160. $f(x) = 3^{4x+1} + 2^{4x+2}$
161. $f(y) = 3^{y^2+1}$ **162.** $f(x) = 2^x$ **163.** $f(x) = 2^{-x}$
164. $f(t) = (\frac{1}{2})^t$ **165.** $f(x) = e^x \ln x$ **166.** $f(x) = \ln(\sin x)$
167. $f(x) = \frac{1}{\ln x}$ **168.** $f(x) = \ln(3xe^{-x})$ **169.** $f(x) = \ln\left(\frac{x-1}{x^2+1}\right)$
170. $f(x) = \ln\left(\frac{e^x}{1+e^x}\right)$
171. $f(x) = \ln(e^{\sin 2x})$ **172.** $f(x) = \ln \sqrt{\frac{x}{x^2+1}}$ **173.** $f(u) = \ln(u^2)$
174. $f(x) = \ln\left(\frac{10}{x}\right)$ **175.** $f(x) = \ln(10^x)$ **176.** $f(x) = \ln(3x) + 4 \ln x$
177. $f(x) = x^2 \ln(2x)$ **178.** $f(x) = \ln(x^{-1})$ **179.** $f(x) = x \ln x$
180. $f(x) = \ln \frac{1}{x}$
181. $f(y) = (\ln y)^3$ **182.** $f(x) = x \ln(\sqrt{x})$ **183.** $f(x) = \ln(7x)$
184. $f(x) = (\ln x)^{1/2}$

J - DIFFERENTIATION

Differentiate the following functions.

- 1.** $f(x) = \sinh(x^2 + 4x)$ **2.** $f(x) = \sinh(7x)$ **3.** $f(x) = \cosh(x^4 + 5x + 7)$
4. $f(t) = \tanh(\sinh 2t)$ **5.** $f(x) = \tanh(x^2 + 5)$ **6.** $f(x) = \frac{\sinh x}{x}$
7. $f(y) = \tanh(2y - y^3)$ **8.** $f(x) = 3 \sinh(x^2) + 2 \cosh(x) - \tanh(3)$
9. $f(x) = 3 \sinh 4x - 2 \cosh 5x$ **10.** $f(x) = \cosh(\tanh x)$
11. $f(x) = \cosh\left(\frac{1}{\sqrt{x}}\right)$ **12.** $f(x) = \tanh\left(\frac{1}{x}\right)$ **13.** $f(t) = \sinh(-t)$
14. $f(x) = \frac{\cosh x}{1+x}$ **15.** $f(x) = 4 \cosh^2(3x)$ **16.** $f(x) = \sinh^2(x^4)$

17. $f(x) = \ln(e^{\cosh 2x})$ 18. $f(\theta) = \arcsin(2\theta)$ 19. $f(x) = \arccos(x^2)$
 20. $f(x) = x(\arcsin x)$
 21. $f(x) = \arcsin\left(\frac{2}{x}\right)$ 22. $f(y) = \arcsin(2y - 3)$ 23. $f(x) = 2x(\arctan x)$
 24. $f(x) = \arctan(5x)$ 25. $f(t) = \arcsin(\sin t)$ 26. $f(x) = \arctan(3x - 4)$
 27. $f(x) = \arccos\left(\frac{x}{4}\right)$

K - IMPLICIT DIFFERENTIATION

The following equations define y implicitly as a function of x .

Determine $\frac{dy}{dx}$ in terms of x and y in each case.

1. $x^2 + y^2 = 100$ 2. $x^3 - y^3 = 6xy$ 3. $x^2y + 3xy^3 - x = 3$ 4. $x^3y^2 - 5x^2y + x = 1$
 5. $x^2 = \frac{x+y}{x-y}$ 6. $\sqrt{xy} + 1 = y$ 7. $(x^2 + 3y^2)^{35} = x$ 8. $\cos xy = y$
 9. $\sin(x^2y^2) = x$ 10. $\tan^3(xy^2 + y) = x$ 11. $\sqrt{3 + \tan xy} - 2 = 0$

L - ANTIDERIVATIVES

Find the most general form of the function f satisfying the following.

1. $f'(x) = x^8$ 2. $f'(x) = \frac{1}{x^6}$ 3. $f'(x) = x^{5/7}$
 4. $f'(\mathbf{x}) = \frac{4}{\sqrt{x}}$ 5. $f'(\mathbf{x}) = \frac{1}{2x^3}$ 6. $f'(\mathbf{x}) = \mathbf{x}^3\sqrt{x}$
 7. $f'(t) = \sqrt[3]{t^2}$ 8. $f'(\mathbf{x}) = (\mathbf{x}^3 - 2\mathbf{x} + 7)$ 9. $f'(\mathbf{x}) = (\mathbf{x}^{-3} + \sqrt{x} - 3\mathbf{x}^{1/4} + \mathbf{x}^2)$
 10. $f'(u) = (u^{2/3} - 4u^{1/5} + 4)$ 11. $f'(x) = \left(\frac{7}{x^{3/4}} - \sqrt[3]{x} + 4\sqrt{x}\right)$
 12. $f'(x) = x(1 + x^3)$ 13. $f'(t) = (1 + t^2)(2 - t)$ 14. $f'(x) = x^{1/3}(2 - x)^2$
 15. $f'(x) = \left[\frac{1}{x^2} - \cos x\right]$ 16. $f'(\theta) = [4\sin \theta + 2\cos \theta]$
 17. $f'(y) = e^y$ 18. $f'(x) = xe^{x^2}$ 19. $f'(x) = \frac{2}{x}$

M - DIFFERENTIAL EQUATIONS

Find the solution of each of the following differential equations with initial values

(This means you have to find the function y .)

1. $\frac{dy}{dx} = \frac{x}{2}, y(\frac{1}{2}) = -1$
2. $\frac{dy}{dx} = -\frac{3}{2}x^2, y(-1) = -\frac{1}{2}$
3. $\frac{dy}{dt} = \sin t, y(\frac{\pi}{2}) = 3$
4. $\frac{dy}{dx} = e^x, y(0) = 4$
5. $\frac{dy}{dx} = \frac{1}{x}, y(1) = 2$
6. $\frac{d^2y}{dx^2} = 0, \frac{dy(2)}{dx} = 1, y(2) = 2$
7. $\frac{d^2y}{dx^2} = \cos x, \frac{dy(0)}{dx} = 1, y(0) = 2$
8. $\frac{d^2y}{dt^2} = e^t, \frac{dy(1)}{dt} = e, y(1) = -4e$

N - INTEGRATION BY INSPECTION

Find the following integrals:

1. $\int e^{5x} dx$
2. $\int x^8 dx$
3. $\int \frac{1}{x^6} dx$
4. $\int t^{5/7} dt$
5. $\int \sqrt[3]{x^2} dx$
6. $\int \frac{4}{\sqrt{x}} dx$
7. $\int \frac{1}{2\theta^3} d\theta$
8. $\int x^3 \sqrt{x} dx$
9. $\int (x^3 - 2x + 7) dx$
10. $\int (x^{-3} + \sqrt{x} - 3x^{1/4} + x^2) dx$
11. $\int (y^{2/3} - 4y^{1/5} + 4) dy$
12. $\int \left(\frac{7}{x^{3/4}} - \sqrt[3]{x} + 4\sqrt{x} \right) dx$
13. $\int x(1 + x^3) dx$
14. $\int (1 + u^2)(2 - u) du$
15. $\int x^{1/3}(2 - x)^2 dx$
16. $\int \left[\frac{1}{t^2} - \cos t \right] dt$
17. $\int [3 \sin x - 4 \cos x] dx$
18. $\int e^{-x} dx$
19. $\int 2xe^{x^2} dx$
20. $\int \frac{1}{2x} dx$
21. $\int e^{3t} dt$
22. $\int \frac{3}{x} dx$
23. $\int \frac{1}{2x+1} dx$
24. $\int \sinh 4x dx$
25. $\int \sinh (4x+6) dx$
26. $\int \cosh (2x+3) dx$
27. $\int \tanh 3x dx$
28. $\int \frac{u^2}{u^3-4} du$
29. $\int \frac{5u^4}{u^5+1} du$
30. $\int \frac{t+1}{t} dt$
31. $\int \frac{x^3}{x^2+1} dx$
32. $\int x^2(2-x^3)^4 dx$
33. $\int xe^{2x^2} dx$
34. $\int \sin \theta (\cos \theta - 3)^3 d\theta$
35. $\int \frac{\sin \sqrt{x}}{\sqrt{x}} dx$
36. $\int \frac{\sin \theta}{1 + \cos \theta} d\theta$
37. $\int \tan \theta d\theta$
38. $\int \frac{1}{x \ln x} dx$
39. $\int \frac{\ln y}{y} dy$
40. $\int (x+1)\sqrt{x^2+2x} dx$